

Question Paper Code : AL3452 — Predicted 2026

B.E./B.Tech. Degree Examinations, November/December 2026

Fourth Semester · Artificial Intelligence and Data Science

AL 3452 — Operating Systems

(Common to Computer Science and Business Systems) · Regulations 2021

Time : Three hours

Maximum : 100 marks

Answer ALL questions · PART A — ($10 \times 2 = 20$ marks) · PART B — ($5 \times 13 = 65$ marks) · PART C — ($1 \times 15 = 15$ marks)

Part A — ($10 \times 2 = 20$ marks)

Unit I — OS Overview & System Calls

1. List any two objectives and two functions of the Operating System. (2)
2. What is a system call? Name the system call used to create processes on UNIX systems. (2)

Unit II — Process Management

3. Define Process Control Block (PCB). Draw the life cycle (state diagram) of a process. (2)
4. State any two differences between user thread and kernel thread. Name the four necessary conditions for deadlock. (2)

Unit III — Memory Management

5. What is Demand Paging? List its advantages. (2)
6. Define Thrashing. What is swapping? (2)

Unit IV — File Systems & I/O

7. State Rotational Latency. Give the role of the OS in free space management. (2)
8. Distinguish between sharable and dedicated devices. List the various file access methods. (2)

Unit V — Virtualization & Mobile OS

9. Define para-virtualization. What are the benefits of virtual machines? (2)
10. Name the mechanism that Android uses to discover and identify activities and services. List any two components unique to mobile OS. (2)

Part B — ($5 \times 13 = 65$ marks)

11. (a)

Very High — 92/100

Unit I — OS Functions, Structure & System Calls

- (i) List down the objectives and functions of Operating Systems in detail. (6)
- (ii) Detail the various types of user interfaces supported by Operating Systems with neat diagrams. (7)

Or

(b)

Very High — 90/100

- (i) Explain the various structures of an Operating System (Monolithic, Layered, Microkernel, Modular) with neat diagrams. (6)
- (ii) Explain the purpose and importance of system calls in detail with examples. Classify the types of system calls. (7)

12. (a) Very High — 95/100

Unit II — CPU Scheduling (appears in ALL 3 papers)

- (i) With a neat sketch, explain the different states of a process. (5)
- (ii) Consider the following set of processes arriving at time 0: (8)

Process	Burst Time (ms)	Priority
P1	6	2
P2	2	4
P3	8	1
P4	3	3
P5	4	2

- (1) Draw the Gantt chart using FCFS, SJF, and Non-preemptive Priority scheduling.
- (2) Compute turnaround time and waiting time for each process.
- (3) Which algorithm gives the minimum average waiting time?

Or

- (b) Very High — 93/100
- (i) What are semaphores? How do they implement mutual exclusion? Explain with the Producer–Consumer problem. (6)
- (ii) An operating system uses the Banker's Algorithm for deadlock avoidance with 3 processes A, B, C and 3 resource types X, Y, Z. Available = (3, 3, 2). (7)

Process	Alloc X	Alloc Y	Alloc Z	Max X	Max Y	Max Z
A	0	1	0	7	5	3
B	2	0	0	3	2	2
C	3	0	2	9	0	2

- (1) Compute the Need matrix.
- (2) Find the safety sequence.
- (3) If process B requests (1,0,2), can the request be granted?

13. (a) Very High — 92/100

Unit III — Memory Management & Page Replacement

- (i) Explain the concept of Paging and its address translation mechanism with neat diagrams. Distinguish between static and dynamic partitions of memory. (7)
- (ii) Given memory partitions of 100K, 400K, 200K, 300K, and 600K. How would First-fit, Best-fit, and Worst-fit algorithms place processes of 112K, 317K, 212K, and 126K? Which algorithm makes the most efficient use of memory? (6)

Or

- (b) Very High — 94/100
- (i) Explain the need and concept of the paging technique in memory management with address translation. (7)
- (ii) Consider the page reference string: **1 2 3 4 1 2 5 1 2 3 4 5** and 3 page frames. Find the number of page faults using FIFO, Optimal, and LRU page replacement algorithms. (6)

Note: Page fault table comparison required for all 3 algorithms.

14. (a) Very High — 91/100

Unit IV — File Systems & Directory Structures

Illustrate the logical structure of a file directory in detail. Compare single-level, two-level, tree-structured, and acyclic-graph directory structures with neat diagrams and examples. (13)

Or

- (b) Very High — 90/100
- (i) Compute the total head movements while servicing requests using FCFS, SCAN, and LOOK disk scheduling algorithms. Consider a disk queue with requests for I/O to disk blocks on cylinders: **68, 173, 41, 102, 15, 134, 55, 67, 90, 182**. The head is initially at cylinder 50. Cylinders numbered 0 to 199. (8)
- (ii) Write a short note on Free Space Management techniques (Bit vector, Linked list, Grouping, Counting). (5)

15. (a) Very High — 90/100

Unit V — Virtual Machines & Mobile OS

Summarize the major types of Virtual Machines and their implementation in detail. Explain para-virtualization and full virtualization with neat diagrams. (13)

Or

- (b) Very High — 92/100
- (i) Explain the Design Goals and the architecture of Android OS with a neat sketch. (7)
- (ii) Compare Android OS and iOS based on architecture, file system, memory management, and security features. (6)

Part C — (1 × 15 = 15 marks)

16. (a) Very High — 93/100

Part C — Big picture / Comparative / Application question (15 marks)

Perform a comparative analysis of Operating Systems — Windows, Linux, Android, and iOS — based on the following features: (15)

- Manufacturer & architecture supported
- File system supported
- Memory management technique used
- User friendliness, security, reliability, and compatibility
- CPU scheduling algorithm used

Or

(b)

High — 82/100

Consider five processes arriving at time 0 with the following CPU burst times: P1=10ms, P2=29ms, P3=3ms, P4=7ms, P5=12ms. Apply FCFS, SJF (non-preemptive), and Round Robin (quantum = 10ms) scheduling algorithms. Draw Gantt charts, compute average waiting time, turnaround time, and identify which algorithm gives the minimum average waiting time. (15)

Pattern analysis summary

Q11 — OS Functions/Structure 3/3 papers ✓

Q12 — CPU Scheduling 3/3 papers ✓

Q13 — Paging / Page faults 3/3 papers ✓

Q14 — File/Disk scheduling 3/3 papers ✓

Q15 — Virtual machines / Android 3/3 papers ✓

Q16 — OS comparison / Big scheduling 3/3 papers ✓

— All the best — · Predictions based on pattern analysis of 3 OS question papers (Apr/May 2023, Nov/Dec 2023, AL3452) ·
Confidence : 93/100